Conditional evidence

Definition: P(A | B) = P(A and B) / P(B), provided P(B) > 0.

The denominator restricts the sample space to the observed event B.

For disjoint causes A and not A, P(B) = P(B | A)P(A) + P(B | not A)P(not A).

This partition assumption is needed to normalize both possible causes.

Bayes odds

Prior odds for A are P(A) / P(not A), when both probabilities are positive.

The likelihood ratio of evidence E is P(E | A) / P(E | not A).

Posterior odds equal prior odds multiplied by the likelihood ratio.

Proof: divide the two Bayes formulas and cancel the common P(E).

A diagnostic example

A condition affects 1 percent of a population.

A test has sensitivity 90 percent and specificity 95 percent.

True positive mass is 0.01 * 0.90 = 0.009.

False positive mass is 0.99 * 0.05 = 0.0495.

Thus P(condition | positive) = 0.009 / 0.0585 = 2 / 13.

The small base rate matters even though the test is usually correct.

Sequential evidence and its limit

For two conditionally independent observations, multiply their likelihood ratios.

Independence must hold conditional on A and separately conditional on not A.

Duplicating the same positive test result does not provide independent evidence.

Exercise: with prior odds 1:9 and independent likelihood ratios 3 and 2, find posterior odds.

Solution: posterior odds are 6:9 = 2:3, so posterior probability is 2 / 5.